



WSQ COURSE SLIDES

# Basic Urban Farming with Hydroponics

Course Code: TGS-2025053916

Version 2.0 · 29 August 2026

Tertiary Infotech Academy Pte Ltd  
UEN 201200696W

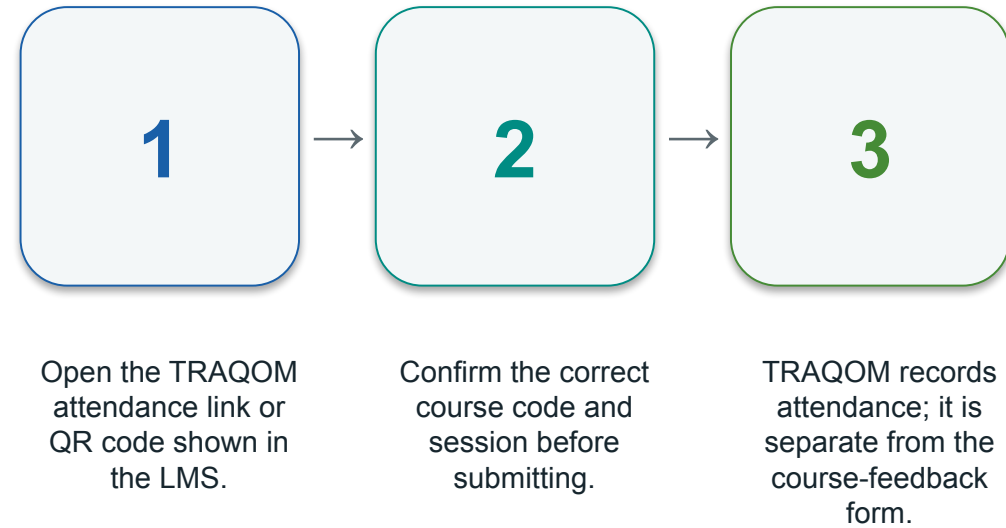
Root-zone control • safe operations • Singapore cases



# Digital Attendance (TRAQOM · SSG)

**ANCHOR: authorised identity → attendance event → confirmation**

- Open the TRAQOM attendance link or QR code shown in the LMS.
- Confirm the correct course code and session before submitting.
- TRAQOM records attendance; it is separate from the course-feedback form.



# Your Trainer



GENERAL TRAINER PROFILE ·  
COMPLETE BEFORE DELIVERY

Name and preferred form of address

Relevant urban-farming / hydroponics experience

Training and Adult Education credentials

Site emergency and escalation role

# Dr Alfred Ang



**COURSE LEAD · TERTIARY  
INFOTECH ACADEMY**

WSQ trainer and courseware lead

Focus: mechanisms, measurements and evidence

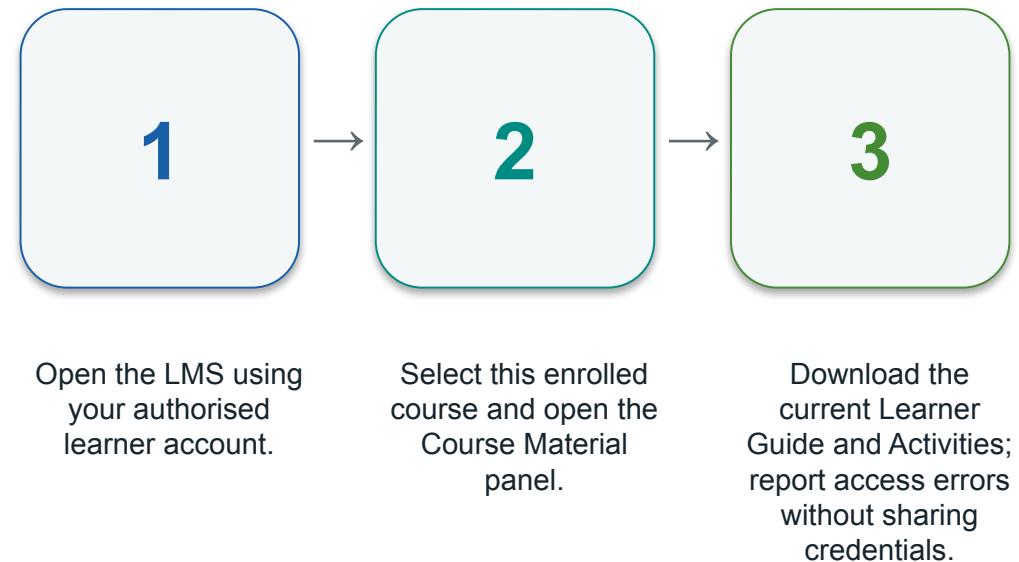
Ask about system selection and troubleshooting

Escalation: assessment and course administration

# Download Course Material

**ANCHOR:** sign in → open enrolled course → download current files

- Open the LMS using your authorised learner account.
- Select this enrolled course and open the Course Material panel.
- Download the current Learner Guide and Activities; report access errors without sharing credentials.



# Learner Introduction

## **ANCHOR: name + role + growing experience + learning goal**

- Share your preferred name and work or community role.
- Describe any hydroponics, gardening, food or facilities experience.
- State one root-zone or system question you want answered today.

**1** Share · your · preferred · name · work · community

**2** Describe · hydroponics · gardening · food · facilities · experience

**3** State · root-zone · system · question · want · answered

# Ground Rules

## **ANCHOR: safe participation + evidence + respect + confidentiality**

- Follow farm, equipment, chemical and electrical safety instructions.
- Ask before touching equipment or photographing a farm area.
- Record facts and label inferences; protect personal and commercial information.
- Keep questions constructive and return shared equipment clean and complete.

**1** Follow farm, equipment, chemical and electrical safety instructions

**2** Ask before touching equipment or photographing a farm area

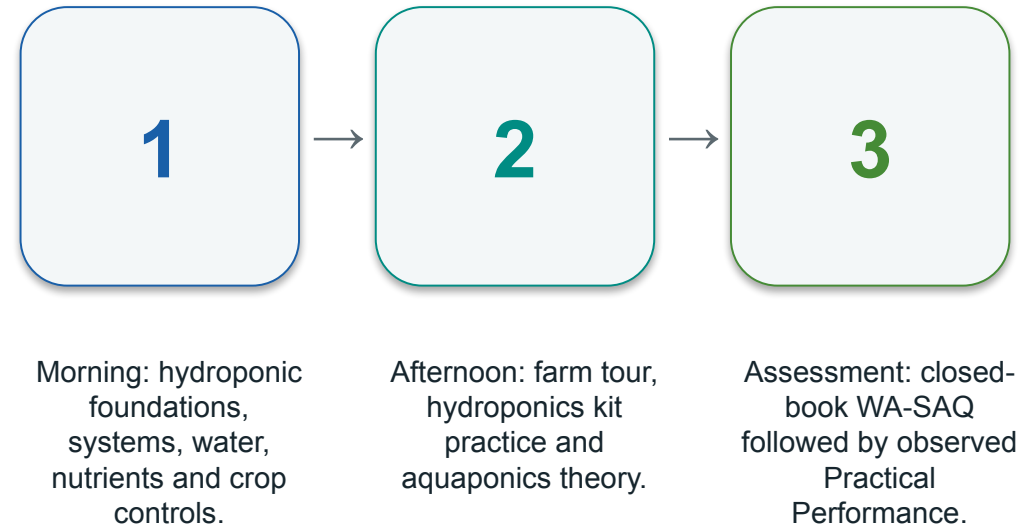
**3** Record facts and label inferences; protect personal and commercial

**4** Keep questions constructive and return shared equipment clean and

# Course Outline

**ANCHOR: root-zone control → systems → production → Singapore cases → activities → aquaponics**

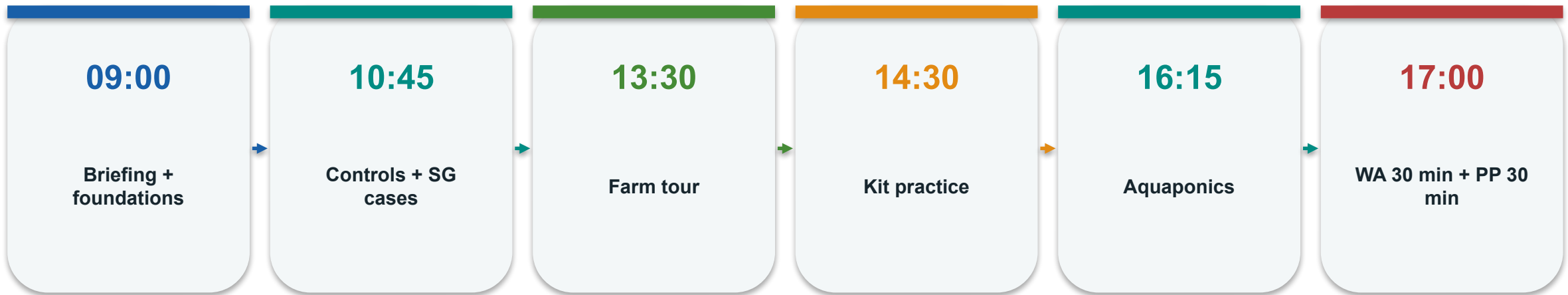
- Morning: hydroponic foundations, systems, water, nutrients and crop controls.
- Afternoon: farm tour, hydroponics kit practice and aquaponics theory.
- Assessment: closed-book WA-SAQ followed by observed Practical Performance.





# Lesson Plan

One-day delivery · 09:00–18:00 · lunch excluded · final assessment 17:00–18:00



Lunch 12:30–13:30 · breaks 10:30–10:45 and 16:00–16:15 · activities use the Farm Tour and Hydroponics Kit checklists.

# Learning Outcomes

## ANCHOR: knowledge K1–K4 + performance A1–A4

- Explain equipment, operating procedures, GAP/WSH and maintenance controls.
- Prepare and operate a hydroponics kit using measured evidence.
- Comply with safety and hygiene requirements; clean, inspect and return equipment.

**1** Explain · equipment · operating · procedures · GAP/WSH · maintenance

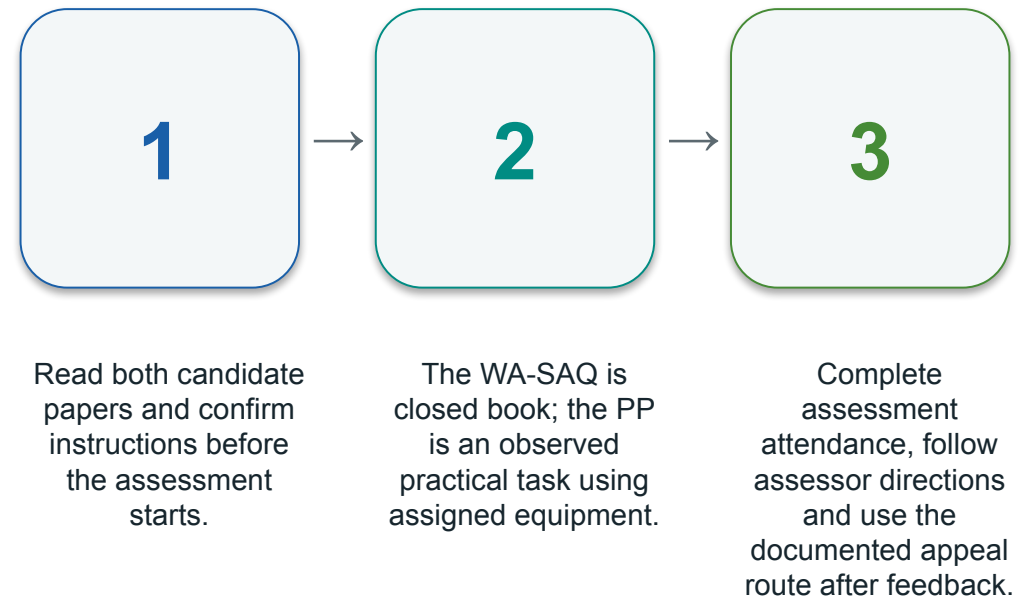
**2** Prepare · operate · hydroponics · using · measured · evidence

**3** Comply · with · safety · hygiene · requirements · clean

# Briefing for Assessment

## **ANCHOR: identity + instructions + permitted resources + evidence + decision**

- Read both candidate papers and confirm instructions before the assessment starts.
- The WA-SAQ is closed book; the PP is an observed practical task using assigned equipment.
- Complete assessment attendance, follow assessor directions and use the documented appeal route after feedback.



# Assessment & Funding

## **ANCHOR: competence evidence is separate from funding administration**

- WA-SAQ: four open-ended questions, closed book, 30 minutes.
- Practical Performance: one observed task covering A1–A4, 30 minutes.
- Attendance, subsidy and claim conditions follow the provider's current SSG administration; confirm any eligibility question with course administration.

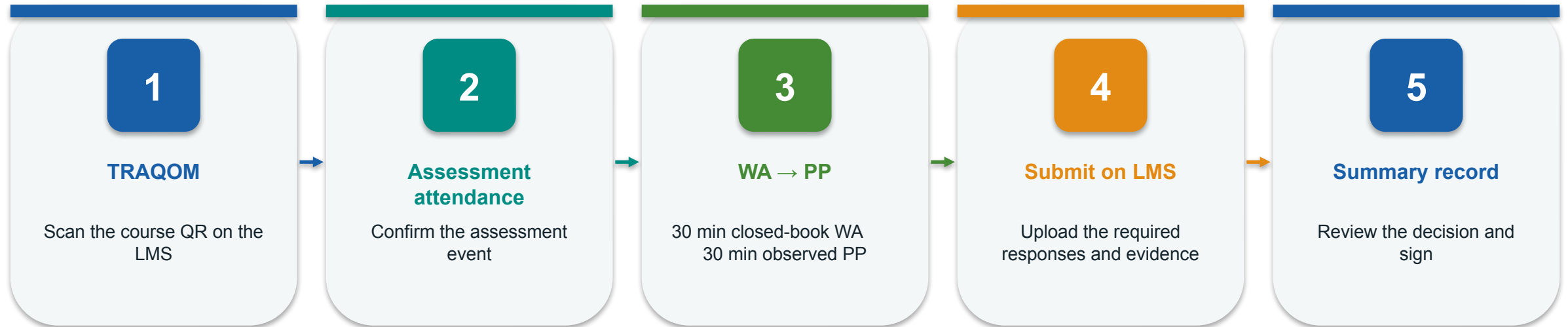
**1** WA-SAQ · four · open-ended · questions · closed · book

**2** Practical · Performance · observed · task · covering · A1–A4

**3** Attendance · subsidy · claim · conditions · follow · provider's

# Assessment Flow

TRAQOM → assessment attendance → closed-book WA → observed PP → LMS submission → signed record



**Assessment sequence: WA-SAQ first, then Practical Performance; follow the candidate instructions and assessor directions.**

# Access the Activities

Use the learner-safe repository for the Farm Tour and Hydroponics Kit activity files

## OPEN ONLINE

Browse the Activities folders and download the required checklist.

## DOWNLOAD ZIP

Use GitHub Code → Download ZIP when a local copy is needed.

<https://github.com/tertiarycourses/TGS-2025053916-Basic-Urban-Farming-with-Hydroponics>

# Practice Exam / Knowledge Check

## **ANCHOR: course WA-SAQ + observed PP are the assessed instruments**

- This is a non-certification WSQ course; revision centres on K1–K4 and A1–A4.
- Use the trainer-provided hydroponics knowledge check and both activity evidence sheets.
- Optional Tertiary practice portal:  
<https://exams.tertiaryinfotech.com/>

**1** This · non-certification · course · revision · centres · K1–K4

**2** trainer-provided · hydroponics · knowledge · check · both · activity

**3** Optional · Tertiary · practice · portal · <https://exams.tertiaryinfotech.com/>

# Safety stop conditions

**ANCHOR:** Stop and notify the trainer when a condition is unsafe or uncontrolled.

- Damaged electrical equipment or water near unprotected power
- Unknown chemical, missing label/SDS/PPE, unstable structure

**1** Damaged · electrical · equipment · water · near · unprotected

**2** Unknown · chemical · missing · label/SDS/PPE · unstable · structure



# Measurement conventions

**ANCHOR: Record value, unit, time, instrument and action.**

- pH is unitless; EC uses mS/cm or  $\mu$ S/cm
- Temperature and calibration status support interpretation

**1**    unitless · uses · mS/cm ·  $\mu$ S/cm

**2**    Temperature · calibration · status · support · interpretation

# 01

## Hydroponic foundations

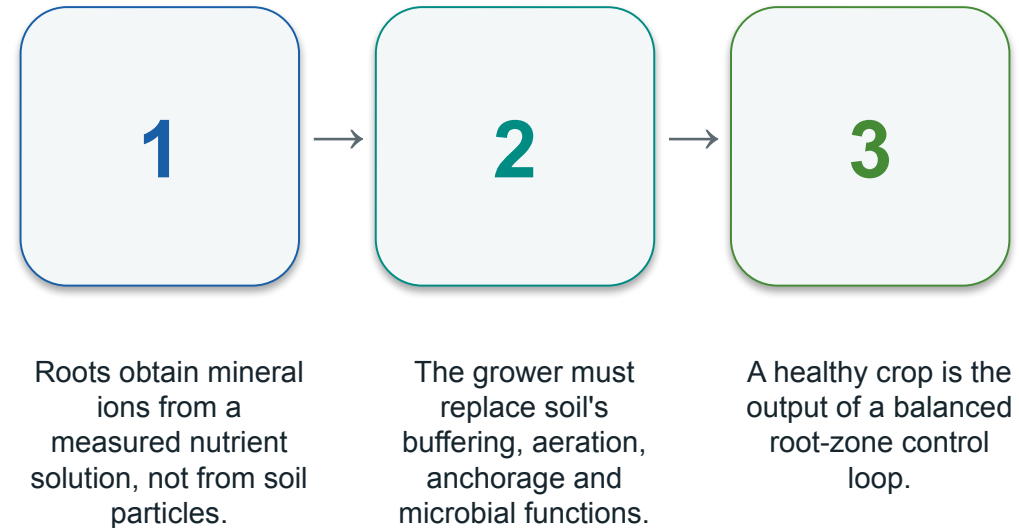
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How water, oxygen, ions and plant demand interact

# Hydroponics is a controlled root-zone system

## ANCHOR: water + oxygen + ions + support

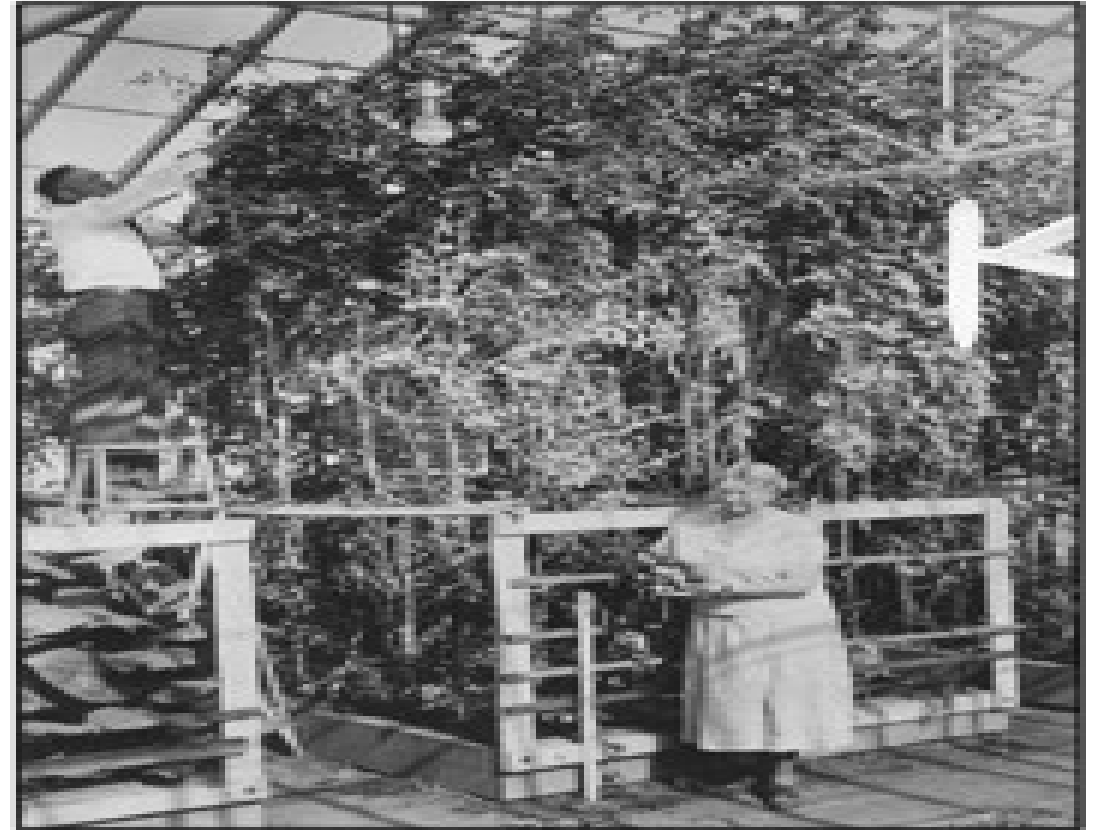
- Roots obtain mineral ions from a measured nutrient solution, not from soil particles.
- The grower must replace soil's buffering, aeration, anchorage and microbial functions.
- A healthy crop is the output of a balanced root-zone control loop.



# The four root-zone variables

## ANCHOR: EC · pH · DO · temperature

- EC estimates total dissolved ionic strength; it does not identify each element.
- pH governs nutrient availability and chemical stability.
- Dissolved oxygen and solution temperature shape respiration and root disease risk.



Legacy visual reused from original slide 22

# Mass balance explains reservoir drift

**ANCHOR:** inputs – plant uptake – discharge = change

- Water loss through transpiration can raise EC even when no fertiliser is added.
- Selective ion uptake means a stable EC can still hide nutrient imbalance.
- Top-up, correction and periodic replacement are different control actions.



Local course photo · EC check



## INPUTS

top-up water + nutrients



## PLANT UPTAKE

water + selective ions



## DISCHARGE

drain + sampling losses



## RESERVOIR Δ

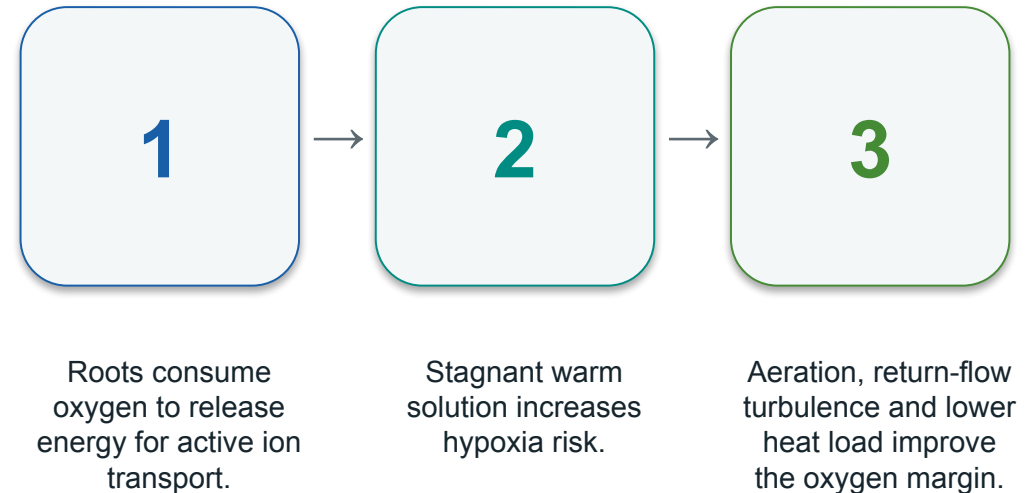
volume + ion profile + EC

Measure trend ≠ diagnose composition

# Root respiration sets an oxygen constraint

**ANCHOR:** warm water holds less dissolved O<sub>2</sub>

- Roots consume oxygen to release energy for active ion transport.
- Stagnant warm solution increases hypoxia risk.
- Aeration, return-flow turbulence and lower heat load improve the oxygen margin.



# Capillary and film flow are not interchangeable

## ANCHOR: water depth changes air-filled porosity

- A thin nutrient film preserves a moist, oxygen-rich root surface.
- Deep flooding provides thermal mass but needs active aeration.
- Media pores can wick water while maintaining structural support.

**1** thin · nutrient · film · preserves · moist · oxygen-rich

**2** Deep · flooding · provides · thermal · mass · needs

**3** Media · pores · wick · water · while · maintaining

# Plant water moves down a potential gradient

**ANCHOR:** root uptake → xylem → stomata

- Light, heat, humidity and airflow change transpiration demand.
- Roots must supply water fast enough to maintain cell turgor.
- Wilting can occur despite a full reservoir when roots are hypoxic or diseased.



Local farm-tour photo · protected crop water-demand context



# Nutrient ions are not plant food

## ANCHOR: photosynthesis supplies carbon energy

- Plants build carbohydrates from  $\text{CO}_2$  and water using light energy.
- Mineral ions enable enzymes, structure and osmotic regulation.
- More fertiliser cannot compensate for inadequate light or damaged roots.



Legacy visual reused from original slide 48

# Soil-free does not mean risk-free

## ANCHOR: one reservoir can connect every plant

- Recirculation conserves water but can spread a waterborne pathogen rapidly.
- Shared chemistry means one dosing error affects the whole crop.
- Monitoring and sanitation are system controls, not optional housekeeping.

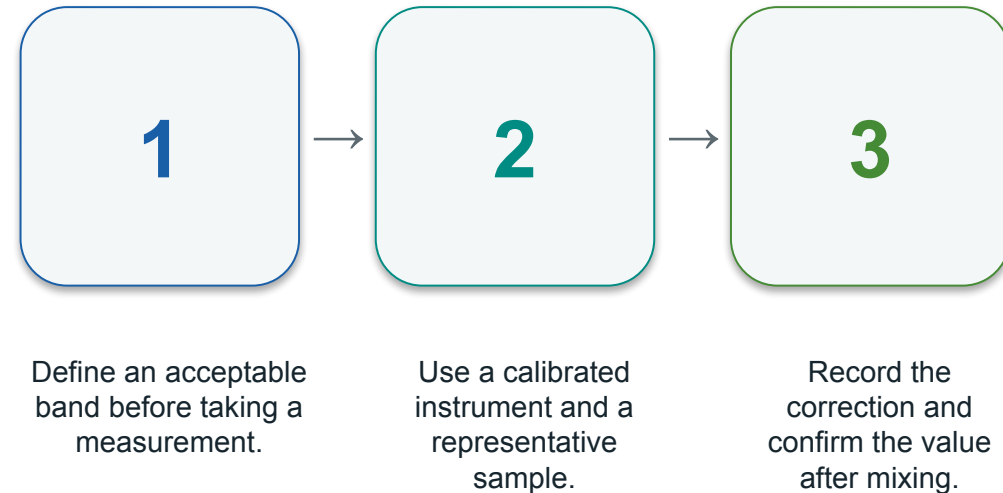


Legacy visual reused from original slide 55

# The production loop is observable

**ANCHOR:** setpoint → measure → compare → act → verify

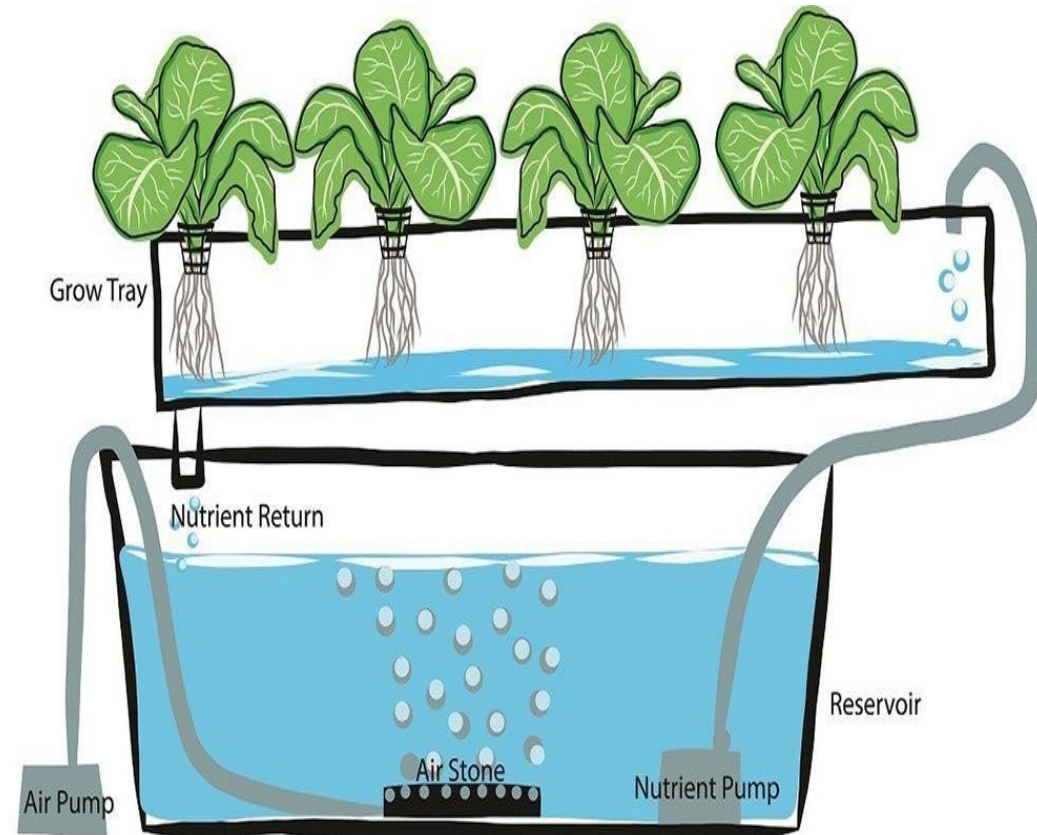
- Define an acceptable band before taking a measurement.
- Use a calibrated instrument and a representative sample.
- Record the correction and confirm the value after mixing.



# Evidence distinguishes control from guesswork

## ANCHOR: time-stamped log + calibration + crop response

- A single reading without time, unit and instrument has weak evidential value.
- Trend data reveals drift before visible symptoms appear.
- Crop photographs should show both canopy and roots, not leaves alone.



Legacy visual reused from original slide 66

# 02

## Hydroponic systems

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Select and diagnose flow, support and failure tolerance

# Nutrient Film Technique (NFT)

## ANCHOR: shallow recirculating film

- Channel slope and flow continuity are critical; loss of pumping exposes roots quickly.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



Legacy visual reused from original slide 75

# Deep Water Culture (DWC)

## ANCHOR: roots suspended in aerated solution

- Air delivery and raft hygiene govern root oxygen and disease risk.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



Legacy visual reused from original slide 80

# Kratky passive culture

## ANCHOR: static reservoir + deliberate air gap

- A growing air gap supplies oxygen; refilling to the original brim can drown roots.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



Legacy visual reused from original slide 87



# Ebb-and-flow

## ANCHOR: timed flood and drain

- Flood duration wets media; complete drainage restores air-filled porosity.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



Legacy visual reused from original slide 91

# Drip hydroponics

## **ANCHOR: emitter uniformity + drainage fraction**

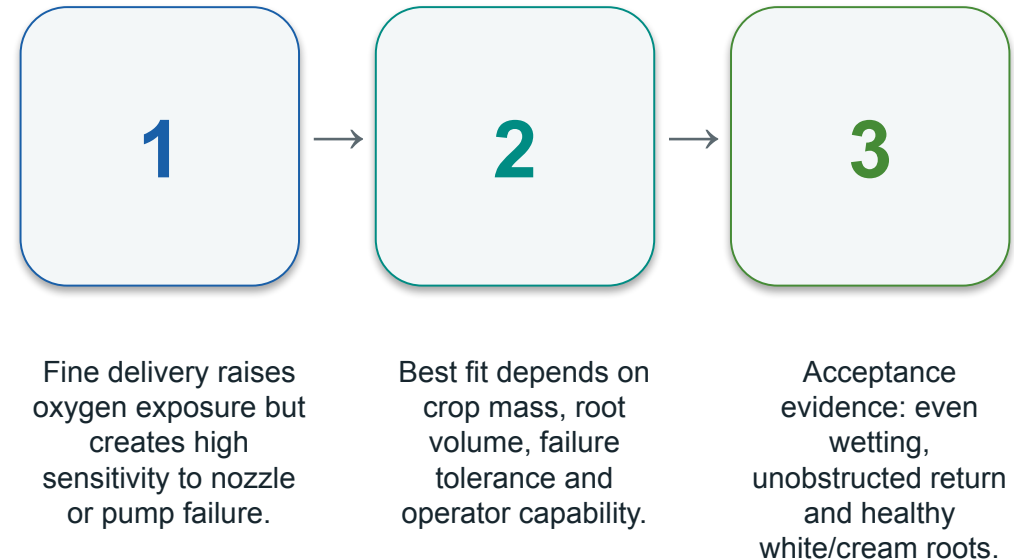
- Emitter blockage and uneven pressure create plant-to-plant variability.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.

**emitter uniformity +  
drainage fraction**

# Aeroponics

## ANCHOR: mist droplet delivery to bare roots

- Fine delivery raises oxygen exposure but creates high sensitivity to nozzle or pump failure.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



# Wick systems

## ANCHOR: passive capillary transport

- Low complexity suits small plants; transport rate limits high-demand crops.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



Legacy visual reused from original slide 110

# Dutch bucket

## **ANCHOR: individual media containers + common return**

- Isolation aids crop support, while a common return retains shared-solution risk.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.

**individual media  
containers + common  
return**

# System selection is a constraint decision

## **ANCHOR: crop × site × utilities × operator**

- Fast leafy greens tolerate different support and cycle times from fruiting crops.
- A rooftop site adds heat, wind, loading and access constraints.
- The simplest system that reliably meets the production target is often the best choice.

**1** Fast · leafy · greens · tolerate · different · support

**2** rooftop · site · adds · heat · wind · loading

**3** simplest · system · that · reliably · meets · production

# Open and recirculating loops trade risk

## **ANCHOR: discharge loss versus shared-reservoir exposure**

- Open drain-to-waste operation simplifies chemistry but loses water and nutrients.
- Recirculation improves resource efficiency and demands stronger monitoring and hygiene.
- Local discharge rules and nutrient recovery should be considered in the SOP.

**1** Open · drain-to-waste · operation · simplifies · chemistry · loses

**2** Recirculation · improves · resource · efficiency · demands · stronger

**3** Local · discharge · rules · nutrient · recovery · should

# Pump duty must include static and friction head

## ANCHOR: duty point = flow at total dynamic head

- A pump's maximum flow is measured near zero head and is not the operating flow.
- Long narrow tubing, elbows and elevation add resistance.
- Measure delivered channel flow instead of relying on the box label.

**1** pump's · maximum · flow · measured · near · zero

**2** Long · narrow · tubing · elbows · elevation · resistance

**3** Measure · delivered · channel · flow · instead · relying



# Return design protects against overflow

## **ANCHOR: inlet flow $\leq$ unobstructed drain capacity**

- Roots, media and biofilm reduce drainage capacity over time.
- A freeboard and emergency overflow limit water damage.
- Power-off testing reveals siphon and reservoir-capacity hazards.

**1**    **Roots · media · biofilm · reduce · drainage · capacity**

**2**    **freeboard · emergency · overflow · limit · water · damage**

**3**    **Power-off · testing · reveals · siphon · reservoir-capacity · hazards**

# Redundancy is proportional to crop loss exposure

## ANCHOR: failure detection time × root drying rate

- NFT may require rapid alarm response because exposed roots dry quickly.
- DWC carries more water reserve but depends on aeration.
- Spare pumps, independent air and tested response actions reduce consequence.

**1** require · rapid · alarm · response · because · exposed

**2** carries · more · water · reserve · depends · aeration

**3** Spare · pumps · independent · tested · response · actions

# Media properties drive irrigation strategy

## ANCHOR: air-filled porosity · water holding · stability

- Coco coir holds water but requires controlled salt management.
- Perlite increases aeration with low mass and low buffering.
- Rockwool is uniform but demands careful wetness control and disposal planning.



Legacy visual reused from original slide 121

# Food-contact materials need verification

## ANCHOR: compatible plastics + cleanable geometry

- Avoid unknown reused containers that may have held chemicals.
- Opaque reservoirs suppress light-driven algae growth.
- Smooth accessible surfaces make cleaning and inspection more reliable.

**1** Avoid · unknown · reused · containers · that · have

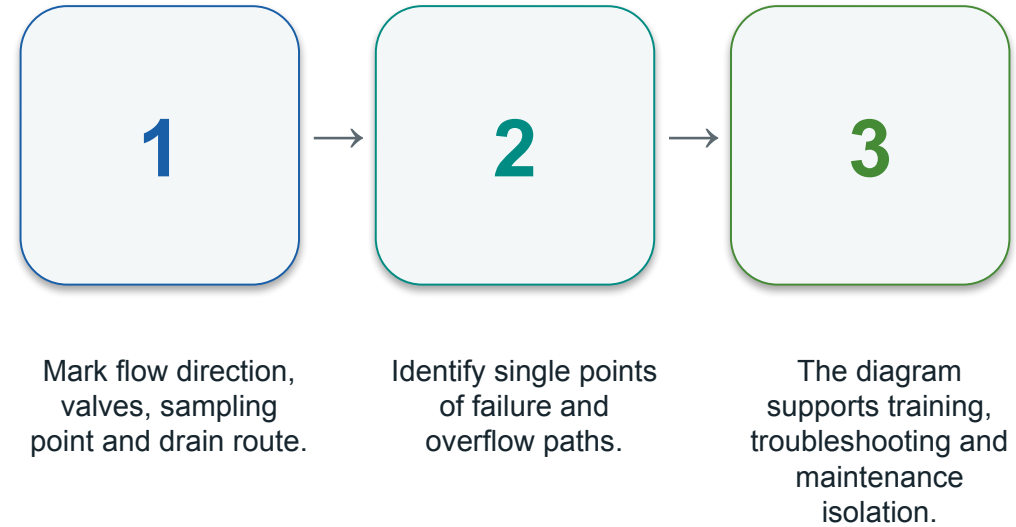
**2** Opaque · reservoirs · suppress · light-driven · algae · growth

**3** Smooth · accessible · surfaces · make · cleaning · inspection

# A system diagram is an operating control

**ANCHOR:** reservoir → pump → crop → return

- Mark flow direction, valves, sampling point and drain route.
- Identify single points of failure and overflow paths.
- The diagram supports training, troubleshooting and maintenance isolation.



# 03

## Water and nutrients

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Measure, mix and correct the root-zone chemistry

# Source water sets the starting chemistry

## ANCHOR: alkalinity differs from pH

- pH is the current hydrogen-ion condition; alkalinity resists pH change.
- High bicarbonate water can drive pH upward after correction.
- Record source EC, pH and where relevant alkalinity before formulating.

**1** current · hydrogen-ion · condition · alkalinity · resists · change

**2** High · bicarbonate · water · drive · upward · after

**3** Record · source · where · relevant · alkalinity · before

# pH controls chemical availability

**ANCHOR:** leafy-green working band commonly ~5.5–6.5

- Outside the operating band, some ions become less available or precipitate.
- Adjust in small increments after nutrients are fully mixed.
- Crop-specific SOPs override a generic band.

**1** Outside · operating · band · some · ions · become

**2** Adjust · small · increments · after · nutrients · fully

**3** Crop-specific · SOPs · override · generic · band



# EC is a proxy, not a recipe

## ANCHOR: mS/cm or $\mu$ S/cm at a stated temperature

- EC increases with dissolved ions from fertiliser and source water.
- Different ion mixtures can produce a similar EC.
- Use the supplier formulation and crop response together with EC.

**1** increases · with · dissolved · ions · from · fertiliser

**2** Different · mixtures · produce · similar

**3** supplier · formulation · crop · response · together · with

# Temperature compensation has limits

## ANCHOR: conductivity changes with temperature

- ATC helps standardise meter readings but does not protect warm roots.
- Record solution temperature with pH and EC.
- Compare trends at similar times of day where possible.

**1** helps · standardise · meter · readings · does · protect

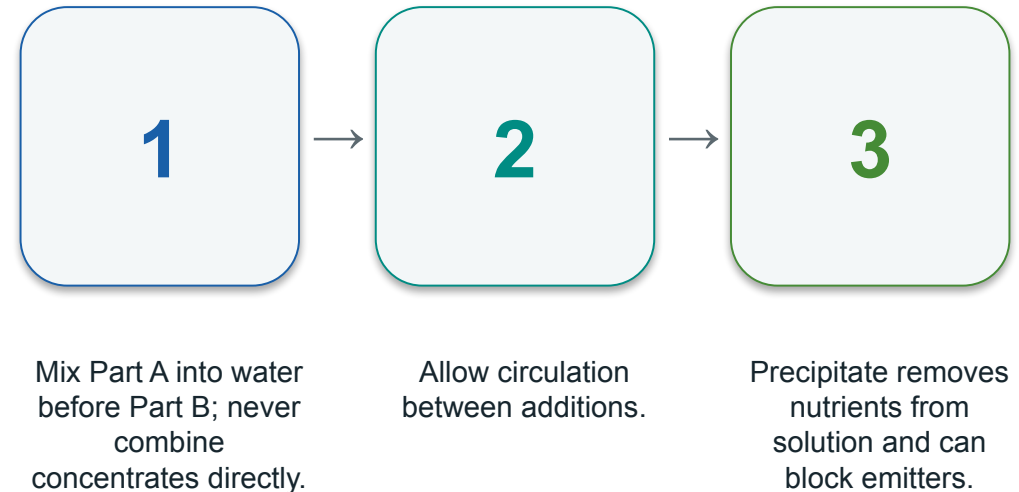
**2** Record · solution · temperature · with

**3** Compare · trends · similar · times · where · possible

# Two-part nutrients prevent concentrate reactions

**ANCHOR:** keep calcium separate from concentrated sulfate/phosphate

- Mix Part A into water before Part B; never combine concentrates directly.
- Allow circulation between additions.
- Precipitate removes nutrients from solution and can block emitters.



# Macronutrients build structure and metabolism

## ANCHOR: N · P · K · Ca · Mg · S

- Nitrogen supports proteins and chlorophyll; imbalance can drive weak lush growth.
- Potassium regulates water balance and enzyme activity.
- Calcium depends strongly on transport to new tissue.

**1** Nitrogen · supports · proteins · chlorophyll · imbalance · drive

**2** Potassium · regulates · water · balance · enzyme · activity

**3** Calcium · depends · strongly · transport · tissue

# Micronutrients operate at small concentrations

**ANCHOR: Fe · Mn · B · Zn · Cu · Mo · Cl · Ni**

- Trace requirement does not mean optional.
- Excess micronutrients can become toxic quickly.
- pH and chelation influence iron availability.

**1** Trace · requirement · does · mean · optional

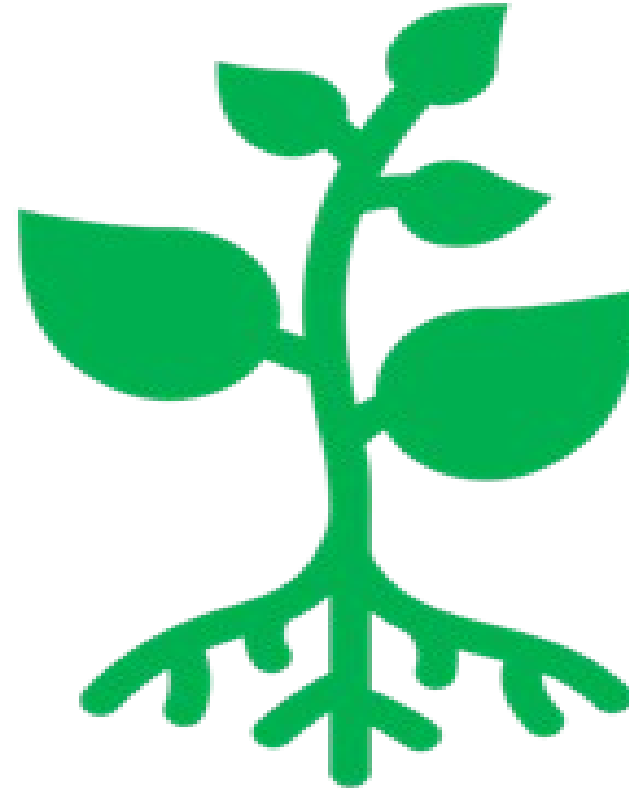
**2** Excess · micronutrients · become · toxic · quickly

**3** chelation · influence · iron · availability

# Symptoms are patterns, not diagnoses

## ANCHOR: old versus new growth + distribution + roots

- Mobile-element shortages often appear first on older leaves.
- Immobile-element disorders often affect new growth.
- Confirm root health, pH, EC and recent operations before correcting nutrients.



Legacy visual reused from original slide 133

# Tip burn can occur with calcium present

## ANCHOR: calcium transport follows transpiration

- Fast inner-leaf growth, high humidity and weak airflow can limit local calcium delivery.
- Raising EC blindly may worsen water uptake.
- Control growth rate, air movement and root conditions.

**1** Fast · inner-leaf · growth · high · humidity · weak

**2** Raising · blindly · worsen · water · uptake

**3** Control · growth · rate · movement · root · conditions

# Iron chlorosis needs differential diagnosis

## ANCHOR: interveinal yellowing on young leaves

- High root-zone pH can reduce iron availability.
- Root damage and cold/wet stress can mimic nutrient shortage.
- Correct the limiting condition before adding more concentrate.

**1** High · root-zone · reduce · iron · availability

**2** Root · damage · cold/wet · stress · mimic · nutrient

**3** Correct · limiting · condition · before · adding · more



# Dilution calculation prevents overdosing

**ANCHOR:**  $C_1 V_1 = C_2 V_2$

- Use consistent units and confirm the concentrate label.
- Estimate conservatively, mix, measure, then trim.
- Wear required PPE and keep acids/bases labelled and segregated.

**1** consistent · units · confirm · concentrate · label

**2** Estimate · conservatively · measure · then · trim

**3** Wear · required · keep · acids/bases · labelled · segregated

# Top-up strategy depends on both water and ions

## ANCHOR: reservoir level + EC trend + crop stage

- Falling level with rising EC suggests water is leaving faster than ions.
- Falling level with falling EC suggests strong nutrient uptake or dilution history.
- Periodic replacement resets selective-ion drift and accumulated residues.

**1** Falling · level · with · rising · suggests · water

**2** Falling · level · with · falling · suggests · strong

**3** Periodic · replacement · resets · selective-ion · drift · accumulated

# Meter calibration is part of the measurement

**ANCHOR:** known standard → rinse → calibrate → verify

- Use fresh, unexpired standards appropriate to the meter.
- Do not pour used standard back into its bottle.
- Record calibration date, standard value and result.

**1** fresh · unexpired · standards · appropriate · meter

**2** pour · used · standard · back · into · bottle

**3** Record · calibration · date · standard · value · result

# Sampling location changes the reading

## **ANCHOR: representative, mixed, bubble-free sample**

- A dosing point can read very differently before complete mixing.
- Rinse the cup and probe with sample solution.
- Consistent sampling improves trend comparability.

**1** dosing · point · read · very · differently · before

**2** Rinse · probe · with · sample · solution

**3** Consistent · sampling · improves · trend · comparability

# pH correction is a controlled chemical task

## ANCHOR: small dose + circulation + remeasurement

- Concentrated acid or alkali can burn skin and roots.
- Add chemical to water as specified; never improvise mixing order.
- Stop if the label, SDS, PPE or correct dosing tool is unavailable.

**1** Concentrated · acid · alkali · burn · skin · roots

**2** chemical · water · specified · never · improvise · mixing

**3** Stop · label · correct · dosing · tool · unavailable

# Reservoir sizing buffers change

## ANCHOR: litres per plant and refill frequency

- A larger volume changes temperature and EC more slowly.
- Excess volume adds weight and cleaning burden.
- Keep enough free capacity for return water during power-off tests.

**1** larger · volume · changes · temperature · more · slowly

**2** Excess · volume · adds · weight · cleaning · burden

**3** Keep · enough · free · capacity · return · water

# Light exclusion is an algae control

**ANCHOR: photons + nutrients + wet surface = algae opportunity**

- Cover reservoirs and unused planting holes.
- Algae competes for oxygen, blocks flow and creates slippery surfaces.
- Clean the root cause rather than relying only on chemical treatment.

1

Cover · reservoirs · unused · planting · holes

2

Algae · competes · oxygen · blocks · flow · creates

3

Clean · root · cause · rather · than · relying

# Biofilm changes hydraulics and hygiene

## **ANCHOR: surface attachment + organic load + time**

- Biofilm can harbour organisms and narrow emitters or channels.
- Inspection targets junctions, dead legs and rough surfaces.
- Cleaning must include physical removal, validated chemistry and rinsing.

**1 Biofilm · harbour · organisms · narrow · emitters · channels**

**2 Inspection · targets · junctions · dead · legs · rough**

**3 Cleaning · must · include · physical · removal · validated**



# 04

## Crop production

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Control nursery, environment, hygiene and crop health

# Crop choice must fit the system envelope

## ANCHOR: cycle time · canopy · root volume · support

- Leafy greens suit short cycles and modest structural support.
- Fruiting crops need longer occupancy, pollination and trellising.
- Marketable yield, not plant count, is the production measure.

**1** Leafy · greens · suit · short · cycles · modest

**2** Fruiting · crops · need · longer · occupancy · pollination

**3** Marketable · yield · plant · count · production · measure

# Seed quality affects the whole batch

## ANCHOR: germination % and uniform emergence

- Record cultivar, lot and sowing date.
- Uniform plugs create more even transplant and harvest timing.
- Separate non-germination from later seedling mortality.

**1** Record · cultivar · sowing · date

**2** Uniform · plugs · create · more · even · transplant

**3** Separate · non-germination · from · later · seedling · mortality

# Nursery moisture balances water and oxygen

## ANCHOR: saturated is not synonymous with healthy

- Overwet media reduces aeration and encourages disease.
- Drying interrupts germination and damages fine roots.
- Use clean water delivery and defined transplant acceptance criteria.

**1** Overwet · media · reduces · aeration · encourages · disease

**2** Drying · interrupts · germination · damages · fine · roots

**3** clean · water · delivery · defined · transplant · acceptance

# Transplant timing is a root-and-canopy decision

## **ANCHOR: established roots + true leaves + intact plug**

- Overgrown seedlings suffer root binding and transition stress.
- Weak seedlings create permanent variability in channel performance.
- Reject diseased or mechanically damaged plugs.

**1** Overgrown · seedlings · suffer · root · binding · transition

**2** Weak · seedlings · create · permanent · variability · channel

**3** Reject · diseased · mechanically · damaged · plugs

# Light quantity is measured as photons

## ANCHOR: PPFD ( $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ ) and DLI

- PPFD describes instantaneous photosynthetic photon flux density.
- DLI integrates PPFD over the photoperiod.
- Uniformity matters because the lowest-light positions limit consistent production.

**1** PPFD · describes · instantaneous · photosynthetic · photon · flux

**2** integrates · PPFD · over · photoperiod

**3** Uniformity · matters · because · lowest-light · positions · limit

# Daily Light Integral links intensity and time

**ANCHOR:  $DLI = PPFD \times \text{seconds} \div 1,000,000$**

- Longer photoperiod can partly compensate for lower intensity within crop limits.
- More photons can raise cooling and electricity demand.
- Measure at canopy height across the growing area.

**1** Longer · photoperiod · partly · compensate · lower · intensity

**2** More · photons · raise · cooling · electricity · demand

**3** Measure · canopy · height · across · growing · area

# Air temperature and humidity shape demand

## ANCHOR: vapour-pressure deficit (VPD)

- Very low evaporative demand can reduce nutrient transport to fast-growing tissue.
- Very high demand can exceed root water supply and close stomata.
- Use crop response with environmental trends, not temperature alone.

**1** Very · evaporative · demand · reduce · nutrient · transport

**2** Very · high · demand · exceed · root · water

**3** crop · response · with · environmental · trends · temperature



# Airflow is a boundary-layer control

## ANCHOR: leaf movement without mechanical stress

- Gentle circulation supports gas exchange and canopy uniformity.
- Dead zones retain humidity and favour disease.
- Excess wind raises water demand and causes abrasion.

**1** Gentle · circulation · supports · exchange · canopy · uniformity

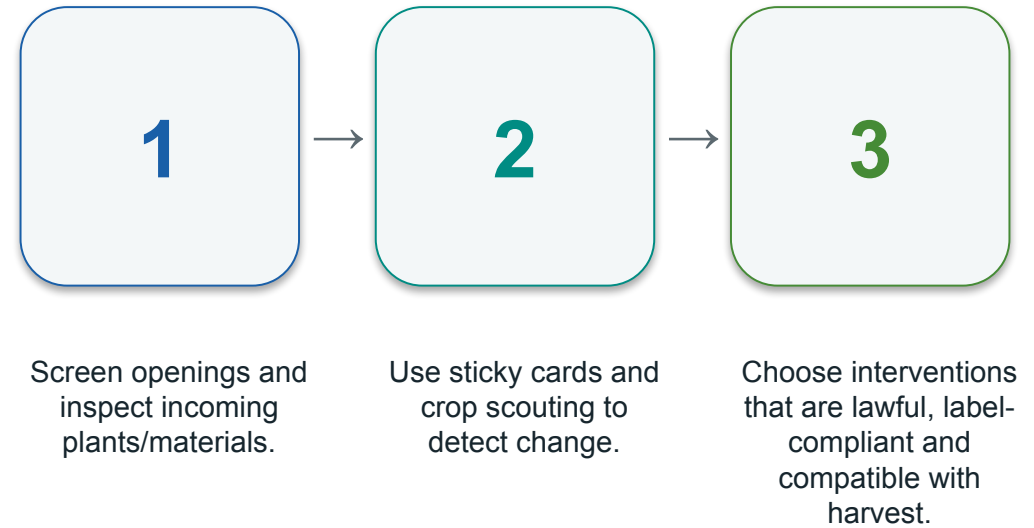
**2** Dead · zones · retain · humidity · favour · disease

**3** Excess · wind · raises · water · demand · causes

# Integrated pest management starts with exclusion

**ANCHOR:** prevent → monitor → identify → intervene → verify

- Screen openings and inspect incoming plants/materials.
- Use sticky cards and crop scouting to detect change.
- Choose interventions that are lawful, label-compliant and compatible with harvest.



# Sticky cards are trend tools

## ANCHOR: standard location + interval + count

- Card colour and placement affect which flying pests are captured.
- Counts are comparable only with a consistent method.
- A trap result must be paired with crop inspection and correct identification.

**1** Card · colour · placement · affect · which · flying

**2** Counts · comparable · only · with · consistent · method

**3** trap · result · must · paired · with · crop

# Root inspection is non-negotiable

## ANCHOR: colour · odour · firmness · branching

- Healthy roots are generally light-coloured and firm, with crop-dependent variation.
- Brown slimy roots and off-odour require rapid investigation.
- Inspect the least vigorous plants and the return end, not only the best canopy.



### HEALTHY SIGNAL

light-coloured · firm · branching



### RAPID INVESTIGATION

brown · slimy · off-odour

Photorealistic root-zone comparison generated for this course

# Harvest hygiene protects ready-to-eat produce

## **ANCHOR: clean hands/tools + cleanable surfaces + lot traceability**

- Separate dirty maintenance tools from harvest equipment.
- Remove damaged or contaminated produce from the marketable lot.
- Record harvest date, crop/lot, quantity and responsible operator.

**1** Separate · dirty · maintenance · tools · from · harvest

**2** Remove · damaged · contaminated · produce · from · marketable

**3** Record · harvest · date · crop/lot · quantity · responsible

# Housekeeping controls slips and contamination

## **ANCHOR: dry walkways + contained chemicals + clear access**

- Leaks are production deviations and safety hazards.
- Electrical connections need dry, protected routing.
- Emergency access, labels and SDS must remain visible.

**1 Leaks · production · deviations · safety · hazards**

**2 Electrical · connections · need · protected · routing**

**3 Emergency · access · labels · must · remain · visible**

# Cleaning is not the same as disinfection

## ANCHOR: soil removal precedes disinfectant contact

- Organic residue shields organisms and consumes disinfectant.
- Correct concentration, contact time and material compatibility are required.
- Rinsing and drying requirements follow the product label/SOP.

**1** Organic · residue · shields · organisms · consumes · disinfectant

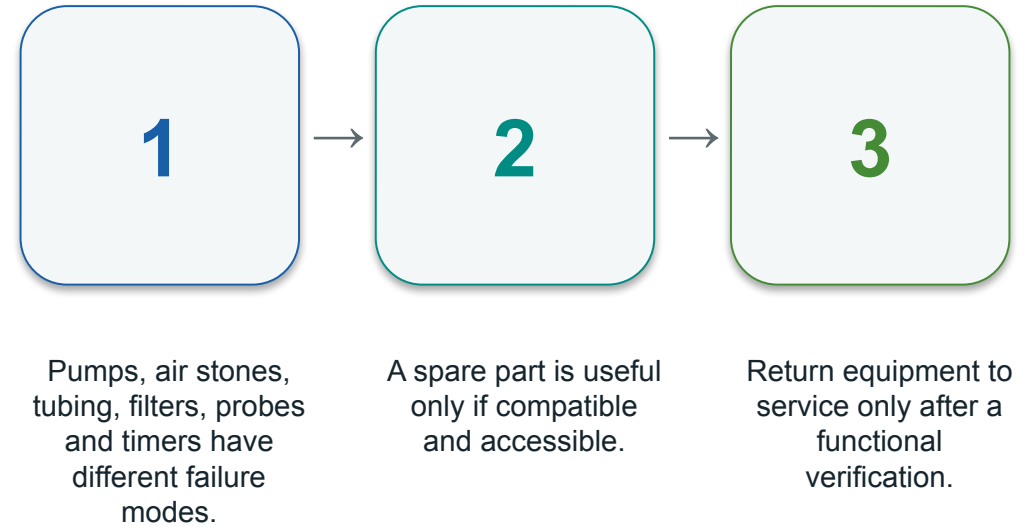
**2** Correct · concentration · contact · time · material · compatibility

**3** Rinsing · drying · requirements · follow · product · label/SOP

# Maintenance preserves control capability

**ANCHOR:** inspect → clean → replace → function-test → record

- Pumps, air stones, tubing, filters, probes and timers have different failure modes.
- A spare part is useful only if compatible and accessible.
- Return equipment to service only after a functional verification.





# 05

## Singapore hydroponics

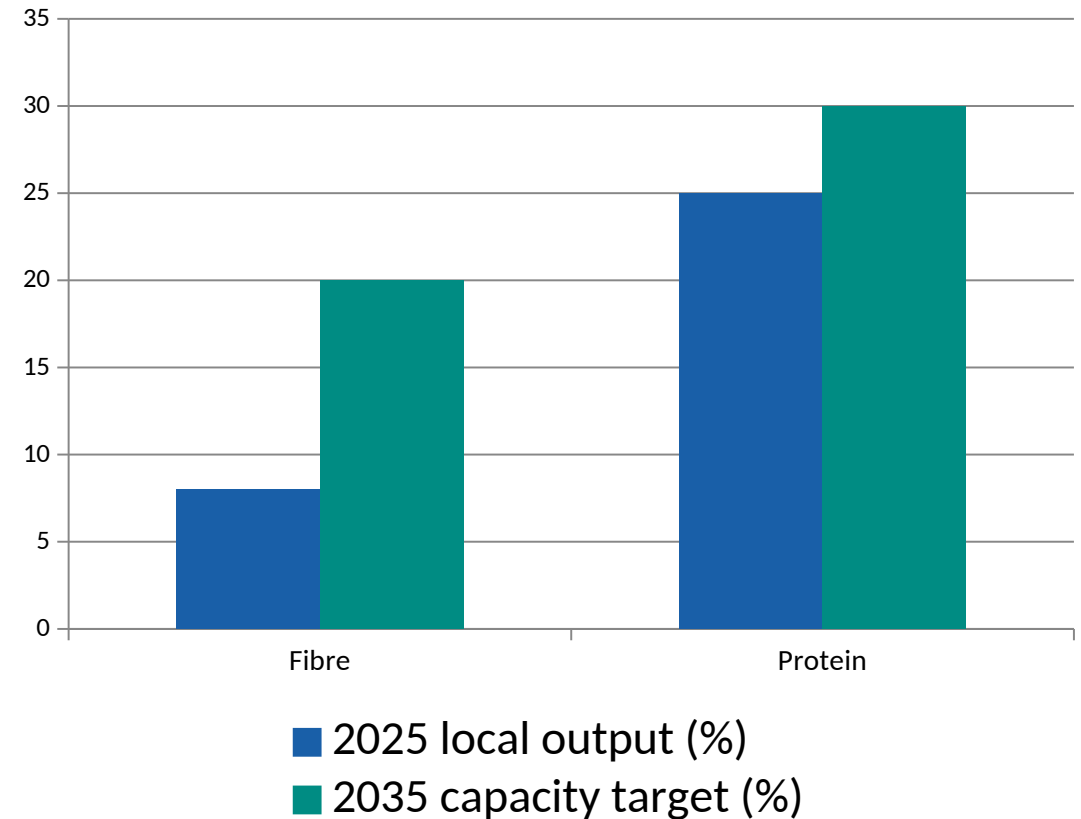
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Current policy, farms, productivity and technology cases

# Singapore Food Story 2 resets the context

## ANCHOR: 2035 capacity targets—20% fibre, 30% protein

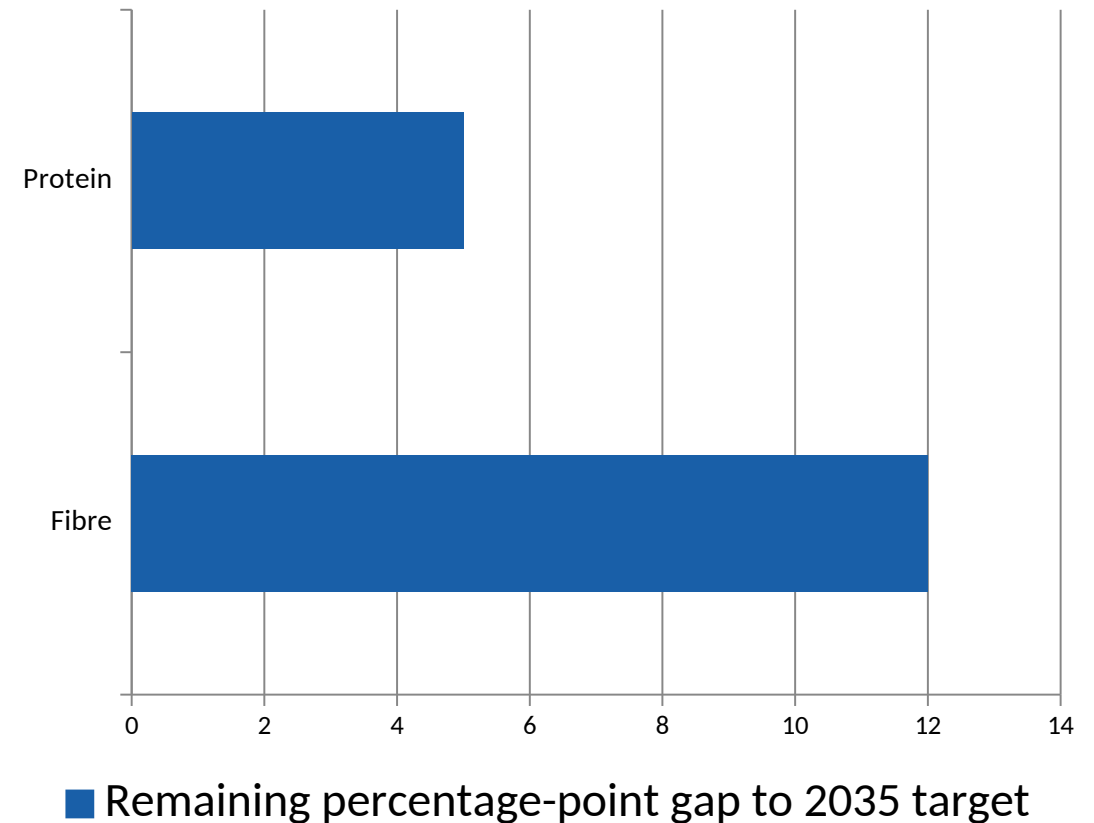
- The current policy focuses on resilient production capacity and commercial viability.
- Previous national framing has been superseded by the current 2035 capacity targets.
- Hydroponics contributes mainly to the fibre category through vegetables.



## 2025 output shows the remaining fibre gap

**ANCHOR: about 8% fibre and 25% protein in 2025**

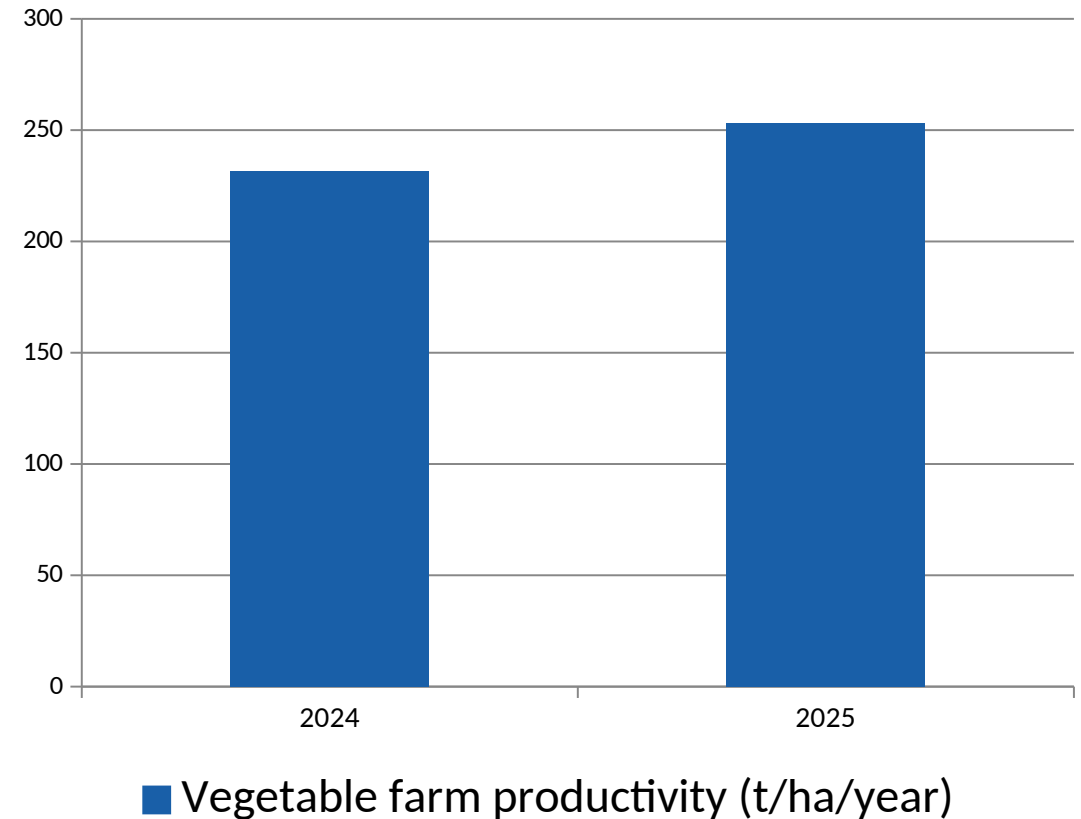
- Local output is only one layer of Singapore's food-resilience strategy.
- Productivity, demand, import diversification and stockpiling interact.
- A technical system must still produce saleable crops at viable cost.



# Vegetable productivity rose in 2025

**ANCHOR: 231.4 → 253.3 tonnes/ha/year**

- SFA reported roughly a 10% increase in farm productivity.
- Area-normalised output supports comparison but does not reveal energy or cost.
- Pair yield with grade-out, input use, labour and downtime.



# ComCrop demonstrates rooftop hybrid production

## ANCHOR: rooftop hydroponics + sunlight + supplemental light

- SFA describes ComCrop as Singapore's first commercial rooftop farm, established in 2011.
- Sensors support monitoring of water, light and nutrient conditions.
- Manual transplanting, harvesting and cleaning remain important operations.



Local farm-tour photo · protected hydroponics (illustrative; not ComCrop)

# Sustenir compresses production into controlled space

## ANCHOR: NFT + vertical layers + controlled environment

- SFA describes recirculating NFT with temperature, humidity and CO<sub>2</sub> control.
- The farm page reports 3.2 tonnes of lettuce from 54 m<sup>2</sup> and over 90 tonnes/year.
- Treat site figures as reported case data, not universal performance.

**1** describes · recirculating · with · temperature · humidity · control

**2** farm · page · reports · tonnes · lettuce · from

**3** Treat · site · figures · reported · case · data

# Singrow targets temperate crops in the tropics

## ANCHOR: cultivar + climate control + automation

- The NUS Enterprise case describes proprietary hydroponic strawberries.
- Computer vision, robotics and crop genetics form one production system.
- Higher-value crops may justify complexity that leafy greens cannot.

**1** Enterprise · case · describes · proprietary · hydroponic · strawberries

**2** Computer · vision · robotics · crop · genetics · form

**3** Higher-value · crops · justify · complexity · that · leafy

# Greenphyto signals a scale-and-automation bet

## ANCHOR: reported 23 m chambers and up to 2,000 t/year

- The Business Times reported an S\$80 million multi-storey hydroponic facility in January 2026.
- The report describes automated seeding, dosing, crop movement and climate control.
- Capacity and company claims must be distinguished from verified realised output.

**1** Business · Times · reported · S\$80 · million · multi-storey

**2** report · describes · automated · seeding · dosing · crop

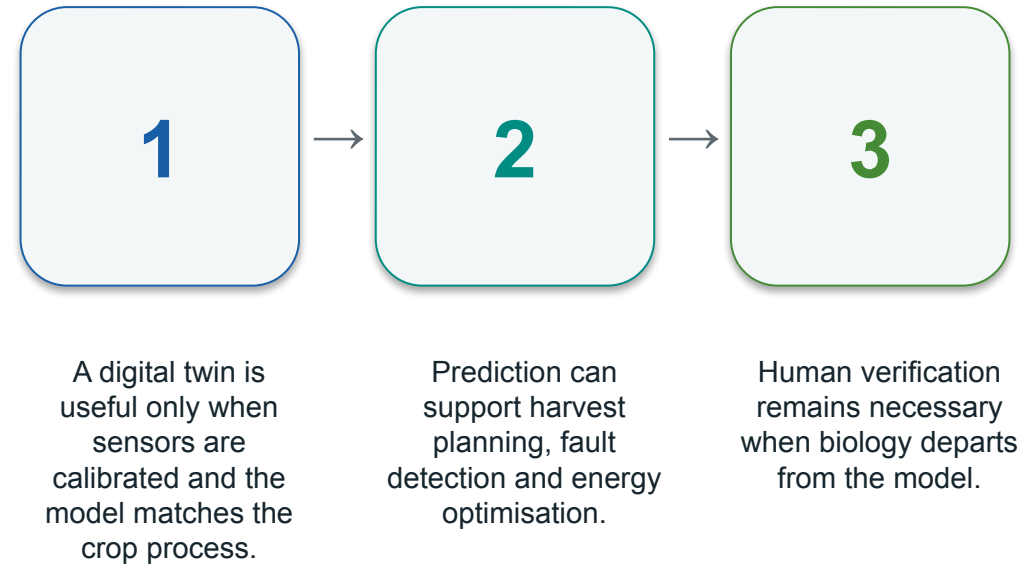
**3** Capacity · company · claims · must · distinguished · from



# Digital twins turn operations into models

**ANCHOR:** sensor state → model → prediction → action

- A digital twin is useful only when sensors are calibrated and the model matches the crop process.
- Prediction can support harvest planning, fault detection and energy optimisation.
- Human verification remains necessary when biology departs from the model.



# Automation changes the job, not the biology

## **ANCHOR: actuators still depend on trustworthy measurements**

- Automated dosing magnifies a bad probe or wrong setpoint.
- Alarms require ownership, escalation and tested response time.
- Operators need crop observation plus technical troubleshooting skills.

**1** Automated · dosing · magnifies · probe · wrong · setpoint

**2** Alarms · require · ownership · escalation · tested · response

**3** Operators · need · crop · observation · plus · technical

# ACT Fund 2 supports capability transformation

## ANCHOR: new S\$70m tranche from 2026

- SFA lists capability upgrading, technology adoption and standards pathways.
- A grant does not remove the need for business and technical feasibility.
- Define measurable production and capability outcomes before procurement.

**1** lists · capability · upgrading · technology · adoption · standards

**2** grant · does · remove · need · business · technical

**3** Define · measurable · production · capability · outcomes · before

# SG GAP connects production with assurance

## **ANCHOR: food safety + quality + environmental + worker welfare controls**

- Good agricultural practice depends on documented, repeatable controls.
- Records support traceability and corrective action.
- Certification scope and current requirements must be checked with SFA/standards sources.

**1** Good · agricultural · practice · depends · documented · repeatable

**2** Records · support · traceability · corrective · action

**3** Certification · scope · current · requirements · must · checked

# Land productivity alone can mislead

**ANCHOR:** yield/m<sup>2</sup> must be read with kWh, labour, water and grade-out

- Vertical stacking can raise output per floor area.
- Lighting, cooling, pumping and dehumidification add operating load.
- A resilient farm optimises the whole system, not a single headline metric.

**1** Vertical · stacking · raise · output · floor · area

**2** Lighting · cooling · pumping ·  
dehumidification · operating · load

**3** resilient · farm · optimises · whole · system ·  
single

# Tropical heat is a root-zone design input

**ANCHOR:** heat gain → warmer solution → lower oxygen margin

- Shade, insulation, reservoir location and operating time influence heat load.
- Cooling can protect crops but adds capital and energy cost.
- Measure the actual daily temperature profile before choosing controls.

**1** Shade · insulation · reservoir · location · operating · time

**2** Cooling · protect · crops · adds · capital · energy

**3** Measure · actual · daily · temperature · profile · before

# Farm economics start with marketable yield

**ANCHOR:** revenue = saleable mass × realised price

- High biological yield can coexist with poor grade-out or weak demand.
- Cycle time and space occupancy affect annual throughput.
- Include packaging, logistics, rejects, energy, labour and downtime.

**1** High · biological · yield · coexist · with · poor

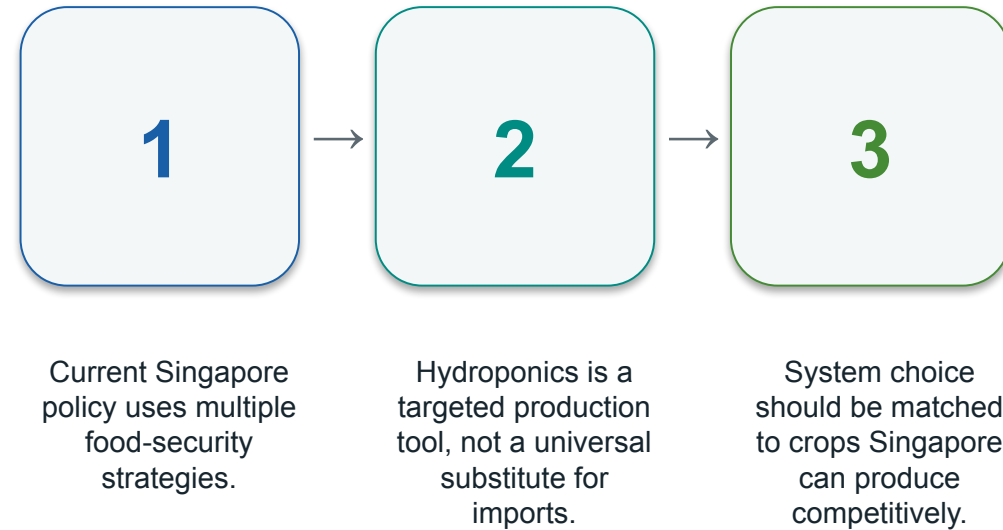
**2** Cycle · time · space · occupancy · affect · annual

**3** Include · packaging · logistics · rejects · energy · labour

# Resilience needs more than local farms

## ANCHOR: local capacity + diversified imports + stockpiles + partnerships

- Current Singapore policy uses multiple food-security strategies.
- Hydroponics is a targeted production tool, not a universal substitute for imports.
- System choice should be matched to crops Singapore can produce competitively.





# 06

## Farm tour activity

---

Observe MEOD Farm using evidence-based checklists

---

# Lunch break

---

Resume at the farm-tour assembly point stated by the trainer

# Observe flow before judging appearance

**ANCHOR:** trace reservoir → delivery → root zone → return

- Identify the system boundary and where water can leave it.
- Locate measurement, dosing, filtration and overflow points.
- Record observations; do not touch equipment without permission.



Visual: generated for this course

# Read the crop as a spatial pattern

## ANCHOR: inlet/middle/return + upper/lower tier

- Uniform symptoms suggest a shared input; local symptoms suggest position-specific causes.
- Compare canopy colour, size and density across the system.
- Photographs require farm permission and must avoid faces or confidential displays.

**1** Uniform · symptoms · suggest · shared · input · local

**2** Compare · canopy · colour · size · density · across

**3** Photographs · require · farm · permission · must · avoid

# Audit water and nutrient controls

## ANCHOR: instrument + setpoint + frequency + response

- Ask where and when pH, EC and temperature are measured.
- Look for calibration, dosing and corrective-action records.
- Distinguish observed evidence from an operator explanation.

**1** where · when · temperature · measured

**2** Look · calibration · dosing · corrective-action · records

**3** Distinguish · observed · evidence · from · operator · explanation

# Audit WSH and food-safety controls

**ANCHOR:** hazard → existing control → residual risk

- Observe wet floors, electrical routing, chemical storage and manual handling.
- Look for clean/dirty zoning and harvest-tool control.
- Escalate urgent hazards to the trainer; never intervene in machinery.

**1** Observe · floors · electrical · routing · chemical · storage

**2** Look · clean/dirty · zoning · harvest-tool · control

**3** Escalate · urgent · hazards · trainer · never · intervene

# Map preventive maintenance

## **ANCHOR: asset ID + interval + completion + function test**

- Check pumps, filters, air delivery, sensors, lighting and structural supports.
- Ask how failure is detected and who responds.
- Look for spare-part readiness and documented return-to-service tests.

**1** Check · pumps · filters · delivery · sensors · lighting

**2** failure · detected · responds

**3** Look · spare-part · readiness · documented · return-to-service · tests

# Evaluate pest management evidence

## **ANCHOR: monitoring location + count trend + intervention record**

- Identify exclusion barriers, scouting routines and sticky-card placement.
- Ask how pests are identified before treatment.
- Confirm interventions respect label, food-safety and pre-harvest requirements.

**1** Identify · exclusion · barriers · scouting · routines · sticky-card

**2** pests · identified · before · treatment

**3** Confirm · interventions · respect · label · food-safety · pre-harvest



# Separate claim, observation and inference

## ANCHOR: C/O/I notation

- Claim: what the guide/operator says.
- Observation: what you directly see, measure or read.
- Inference: your explanation, labelled as provisional until verified.

**1** Claim · what · guide/operator · says

**2** Observation · what · directly · measure · read

**3** Inference · your · explanation · labelled · provisional · until

# Farm-tour evidence is a checklist package

**ANCHOR:** completed PDF + permitted photos + one flow sketch

- The activity PDF defines the required evidence.
- Use concise factual notes with location and time.
- Submit no personal data, trade secrets or unsafe photographs.



**OBSERVE** · crop condition, spatial patterns and controls



**ASK** · verify the operator explanation



**RECORD** · location, time and permission

Tertiary Infotech farm tours · blog archive, 15 Feb 2020

# 07

## Hydroponics kit practice

---

Commission, verify and return a kit to a controlled state

# Kit readiness is an acceptance state

**ANCHOR:** clean · intact · stable · leak-free · electrically safe

- A visual inspection occurs before water or power is applied.
- Tubing must be secured without kinks and returns must be unobstructed.
- Reject damaged cables, unknown containers or unstable structures.

**1** visual · inspection · occurs · before · water · power

**2** Tubing · must · secured · without · kinks · returns

**3** Reject · damaged · cables · unknown · containers · unstable

# Flow verification uses evidence

**ANCHOR: every site wet + return stable + no overflow**

- Observe the last planting site, not only the first.
- Confirm pump noise and vibration are normal for the unit.
- A timed-volume check can quantify delivered flow where required.

**1** Observe · last · planting · site · only · first

**2** Confirm · pump · noise · vibration · normal · unit

**3** timed-volume · check · quantify · delivered · flow · where

# A baseline precedes correction

## ANCHOR: source water pH/EC/temperature recorded first

- Baseline data separates source-water contribution from added nutrients.
- Use clean labelled tools to avoid cross-contamination.
- The Learner Guide contains the detailed sequence and safety checks.

**1** Baseline · data · separates · source-water · contribution · from

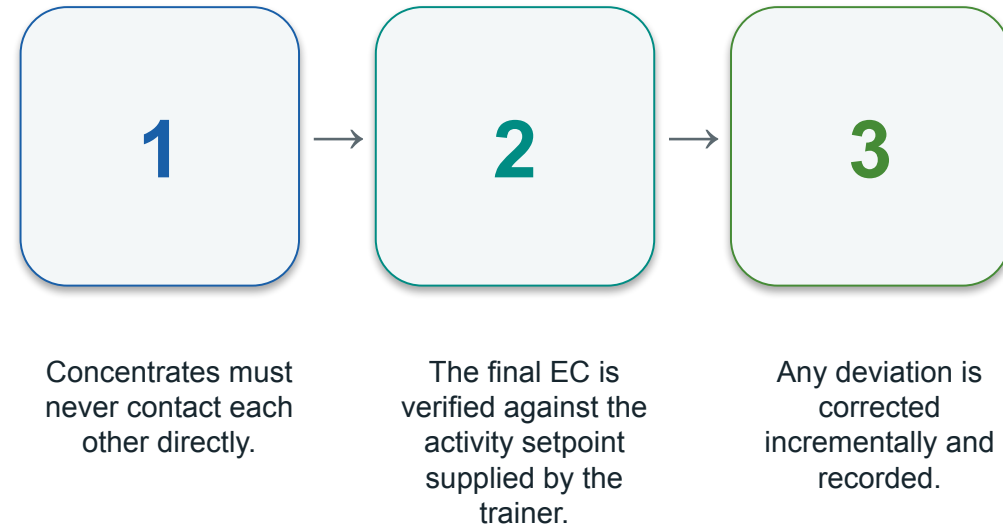
**2** clean · labelled · tools · avoid · cross-contamination

**3** Learner · Guide · contains · detailed · sequence · safety

# Mixing order protects chemistry

**ANCHOR:** water → A → circulate → B → circulate → measure

- Concentrates must never contact each other directly.
- The final EC is verified against the activity setpoint supplied by the trainer.
- Any deviation is corrected incrementally and recorded.



# pH adjustment uses a stop condition

## ANCHOR: no label/SDS/PPE = no dosing

- Use the trainer-provided product and dedicated dosing tool only.
- Add small amounts, circulate and remeasure.
- Never return excess chemical to the stock container.

**1**    **trainer-provided · product · dedicated · dosing  
· tool · only**

**2**    **small · amounts · circulate · remeasure**

**3**    **Never · return · excess · chemical · stock ·  
container**



# Transplant acceptance protects uniformity

## **ANCHOR: intact plug + healthy root + centred support + stable crown**

- Roots should contact moisture without burying the crown.
- Remove weak or diseased seedlings from the production batch.
- Plant labels preserve cultivar and date traceability.

**1**    Roots · should · contact · moisture · without · burying

**2**    Remove · weak · diseased · seedlings · from · production

**3**    Plant · labels · preserve · cultivar · date · traceability

# Power-off behaviour is part of commissioning

## **ANCHOR: no siphon overflow + reservoir contains return volume**

- Simulated power loss exposes drainage and restart behaviour.
- Restart must restore delivery to every site.
- Do not conduct the test where electrical or spill controls are inadequate.

**1** Simulated · power · loss · exposes · drainage · restart

**2** Restart · must · restore · delivery · every · site

**3** conduct · test · where · electrical · spill · controls

# A correction log closes the loop

**ANCHOR:** before → action → after → operator → time

- Record the reason and exact quantity added.
- Allow adequate mixing before the after-reading.
- Escalate if the value does not respond as expected.

**1** Record · reason · exact · quantity · added

**2** Allow · adequate · mixing · before · after-  
reading

**3** Escalate · value · does · respond · expected

# Root-zone diagnosis starts below the canopy

## ANCHOR: roots + flow + temperature + chemistry

- Leaf symptoms alone can lead to the wrong correction.
- Inspect for obstruction, low aeration, warmth and odour.
- Isolate suspect plants and protect the shared reservoir where required by SOP.

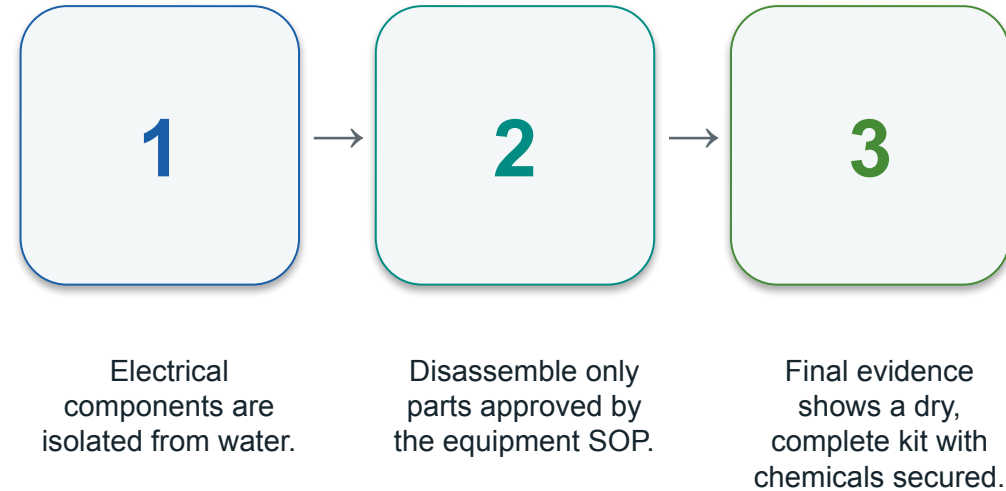


Local farm-tour photo · channel ends and return plumbing

# Cleaning returns the kit to a controlled state

**ANCHOR:** drain → remove residue → clean → rinse → dry  
→ store

- Electrical components are isolated from water.
- Disassemble only parts approved by the equipment SOP.
- Final evidence shows a dry, complete kit with chemicals secured.



# Practice evidence maps to A1–A4

## ANCHOR: prepare · operate · comply · maintain

- A1: materials and equipment ready.
- A2: system operates to target with measured evidence.
- A3: WSH, food safety and housekeeping controls observed.
- A4: cleaning, maintenance check and storage completed.

1

**A1: materials and equipment ready**

2

**A2: system operates to target with measured evidence**

3

**A3: WSH, food safety and housekeeping controls observed**

4

**A4: cleaning, maintenance check and storage completed**

# 08

## Aquaponics theory

---

Balance fish, microbes, plants and water quality



# Aquaponics couples two biological processes

**ANCHOR:** fish waste → nitrification → plant uptake

- Fish produce ammonia-containing waste.
- Nitrifying microbes convert ammonia to nitrite and then nitrate.
- Plants remove nutrients, while water returns to the culture system.



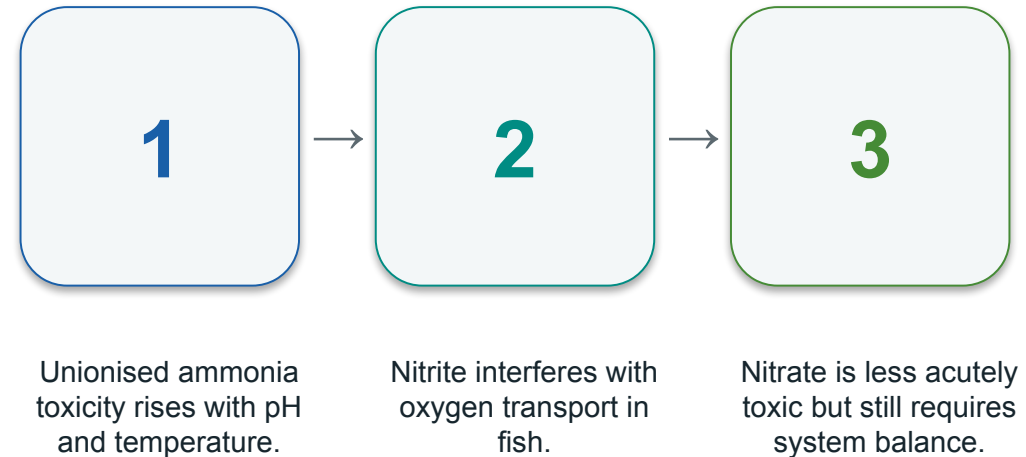
Visual: generated for this course



# The nitrogen cycle has toxic intermediates

**ANCHOR:**  $\text{NH}_3/\text{NH}_4^+ \rightarrow \text{NO}_2^- \rightarrow \text{NO}_3^-$

- Unionised ammonia toxicity rises with pH and temperature.
- Nitrite interferes with oxygen transport in fish.
- Nitrate is less acutely toxic but still requires system balance.



# Three living groups share one water loop

## ANCHOR: fish · microbes · plants

- A setpoint suitable for plants alone may harm fish or biofilter microbes.
- Feeding rate drives nutrient loading and oxygen demand.
- Management aims for an acceptable compromise, not a single-species optimum.

**1** setpoint · suitable · plants · alone · harm · fish

**2** Feeding · rate · drives · nutrient · loading · oxygen

**3** Management · aims · acceptable · compromise · single-species · optimum

# pH is a compromise variable

## ANCHOR: fish health + nitrification + nutrient availability

- Nitrification tends to acidify the system over time.
- Very high pH can increase toxic unionised ammonia fraction.
- Corrections must be gradual and compatible with all organisms.

**1** Nitrification · tends · acidify · system · over · time

**2** Very · high · increase · toxic · unionised · ammonia

**3** Corrections · must · gradual · compatible · with · organisms

# Dissolved oxygen supports fish, roots and biofilter

## ANCHOR: three simultaneous oxygen demands

- Warm water and high feeding increase risk.
- Aeration failure can become an emergency.
- Independent air delivery and alarms may be justified by stock value and response time.

**1** Warm · water · high · feeding · increase · risk

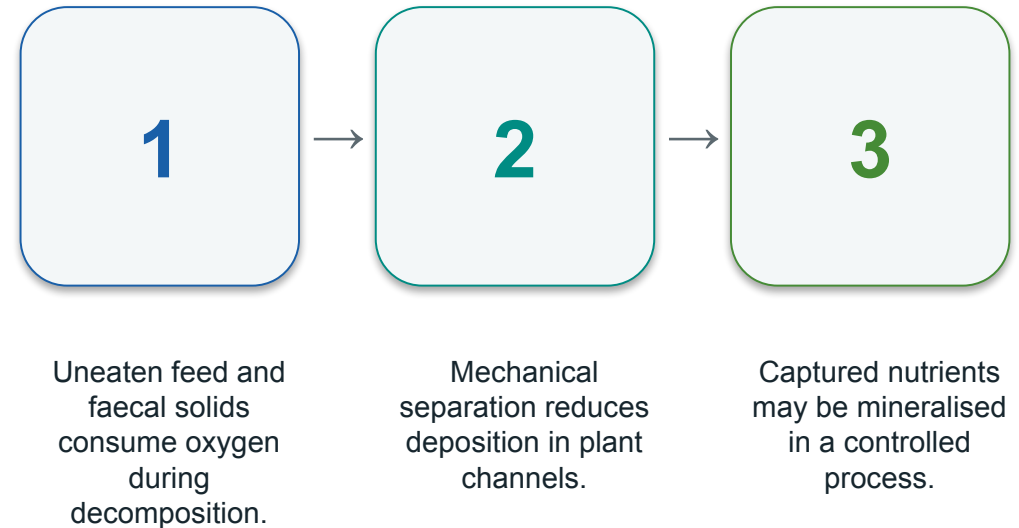
**2** Aeration · failure · become · emergency

**3** Independent · delivery · alarms · justified · stock · value

# Solids management precedes mineralisation

## ANCHOR: settle/filter solids before they foul roots

- Uneaten feed and faecal solids consume oxygen during decomposition.
- Mechanical separation reduces deposition in plant channels.
- Captured nutrients may be mineralised in a controlled process.



# Biofilters need surface area and flow

**ANCHOR: attached nitrifiers + oxygenated water + alkalinity**

- New systems require cycling before full fish loading.
- Disinfectants and sudden chemistry changes can damage the microbial community.
- Monitor ammonia and nitrite during startup and after disturbances.

**1** systems · require · cycling · before · full · fish

**2** Disinfectants · sudden · chemistry · changes · damage · microbial

**3** Monitor · ammonia · nitrite · during · startup · after

# Feeding rate is the nutrient input

## ANCHOR: feed mass/day links fish biomass to plant area

- Overfeeding raises waste, oxygen demand and water-quality risk.
- Underfeeding limits fish growth and plant nutrients.
- Use species-, feed- and system-specific design guidance.

**1** Overfeeding · raises · waste · oxygen · demand · water-quality

**2** Underfeeding · limits · fish · growth · plant · nutrients

**3** species- · feed- · system-specific · design · guidance

# Decoupled aquaponics separates optimisation

## ANCHOR: controlled transfer between fish and plant loops

- A decoupled design can protect fish from crop-specific corrections.
- It adds tanks, controls, monitoring and operating complexity.
- The value depends on production goals and operator capability.

**1** decoupled · design · protect · fish · from · crop-specific

**2** adds · tanks · controls · monitoring · operating · complexity

**3** value · depends · production · goals · operator · capability



# Biosecurity spans water, animals and plants

## **ANCHOR: quarantine + source control + sanitation + records**

- Do not transfer water, nets or plants between systems without control.
- Fish health and food-safety obligations differ from plant-only hydroponics.
- Mortality, treatment and water-quality records support investigation.

**1** transfer · water · nets · plants · between · systems

**2** Fish · health · food-safety · obligations · differ · from

**3** Mortality · treatment · water-quality · records · support · investigation

# Hydroponics and aquaponics solve different constraints

## **ANCHOR: formulated nutrients versus biological nutrient generation**

- Hydroponics offers direct nutrient formulation and easier crop isolation.
- Aquaponics creates coupled biological value and coupled risk.
- Choose based on production objective, market, regulation and operational skill.

**1 Hydroponics · offers · direct · nutrient · formulation · easier**

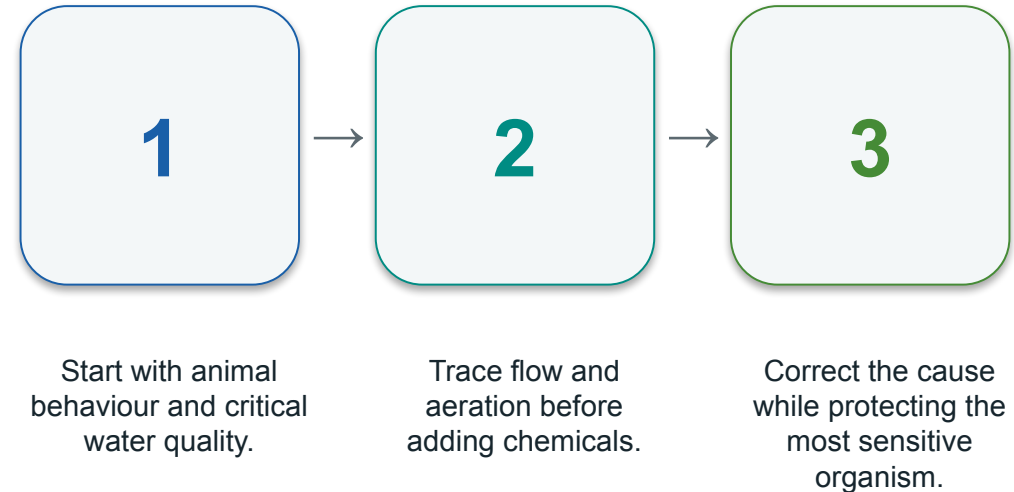
**2 Aquaponics · creates · coupled · biological · value · coupled**

**3 Choose · based · production · objective · market · regulation**

# Aquaponics troubleshooting follows the loop

**ANCHOR:** fish → feed → oxygen → ammonia → biofilter → plant

- Start with animal behaviour and critical water quality.
- Trace flow and aeration before adding chemicals.
- Correct the cause while protecting the most sensitive organism.



# What You Achieved

## **ANCHOR: diagnose the loop + operate safely + verify evidence**

- Explain the water–root–canopy mechanism and select a suitable hydroponic system.
- Measure and interpret pH, EC, temperature, flow and crop/root condition.
- Complete the farm-tour and kit-practice evidence to SOP.

**1 Explain · water–root–canopy · mechanism · select · suitable · hydroponic**

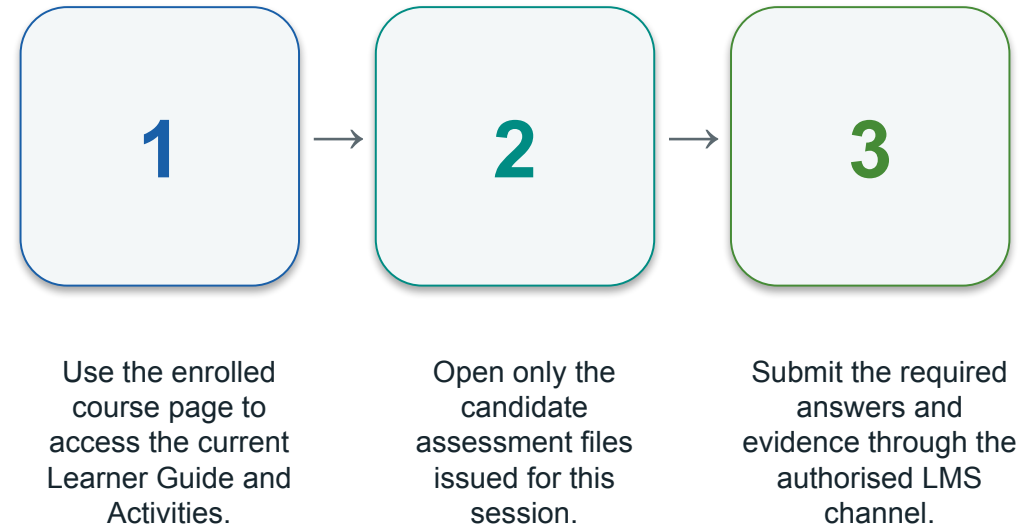
**2 Measure · interpret · temperature · flow · crop/root · condition**

**3 Complete · farm-tour · kit-practice · evidence**

# Courseware & Assessment on the LMS

## **ANCHOR: current files + candidate papers + secure submission**

- Use the enrolled course page to access the current Learner Guide and Activities.
- Open only the candidate assessment files issued for this session.
- Submit the required answers and evidence through the authorised LMS channel.



# Completion & Certification

## **ANCHOR: attendance + completed training + competent assessment decision**

- Complete the required training attendance and both assessment instruments.
- A Competent decision requires the approved WA-SAQ and Practical Performance evidence.
- Eligible competent learners receive the WSQ completion credential through the provider's current SSG process.

**1 Complete · required · training · attendance · both · assessment**

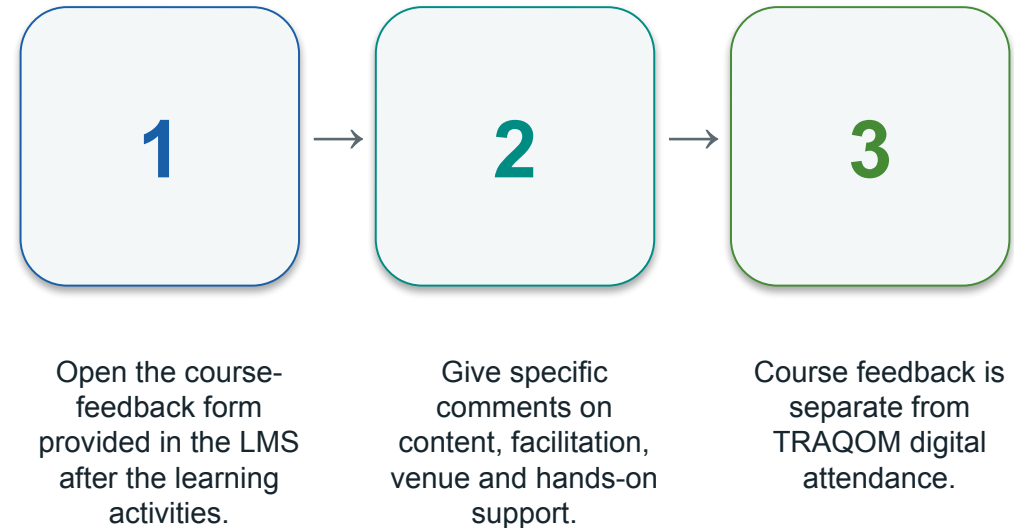
**2 Competent · decision · requires · approved · WA-SAQ · Practical**

**3 Eligible · competent · learners · receive · completion · credential**

# Course Feedback

**ANCHOR:** reflect → rate → comment → submit

- Open the course-feedback form provided in the LMS after the learning activities.
- Give specific comments on content, facilitation, venue and hands-on support.
- Course feedback is separate from TRAQOM digital attendance.



# Assessment

## **ANCHOR: WA-SAQ 30 min → Practical Performance 30 min**

- Complete the assessment digital-attendance event before starting.
- Sit the closed-book WA-SAQ first, then the observed Practical Performance.
- Submit through the LMS and sign the Assessment Summary Record after feedback.

**1** Complete · assessment · digital-attendance · event · before · starting

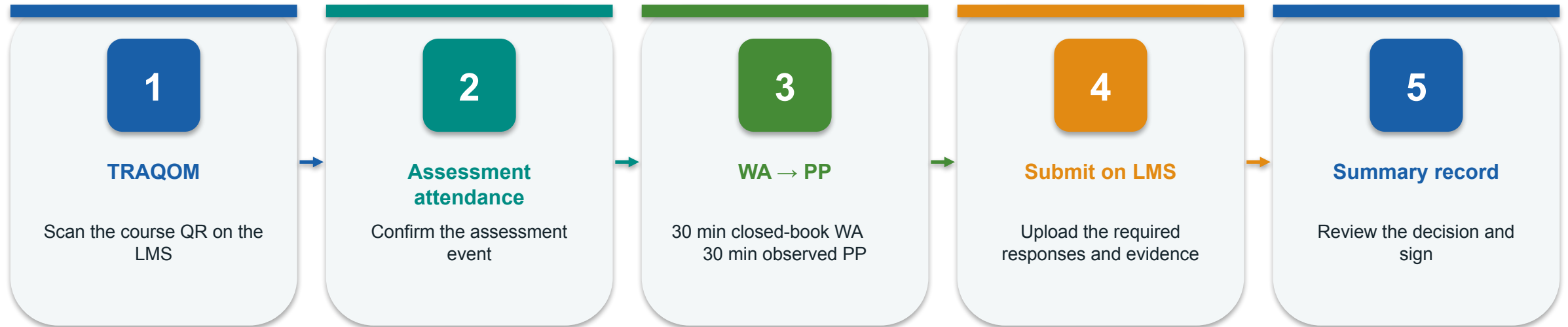
**2** closed-book · WA-SAQ · first · then · observed · Practical

**3** Submit · through · sign · Assessment · Summary · Record



# Assessment Flow

TRAQOM → assessment attendance → closed-book WA → observed PP → LMS submission → signed record

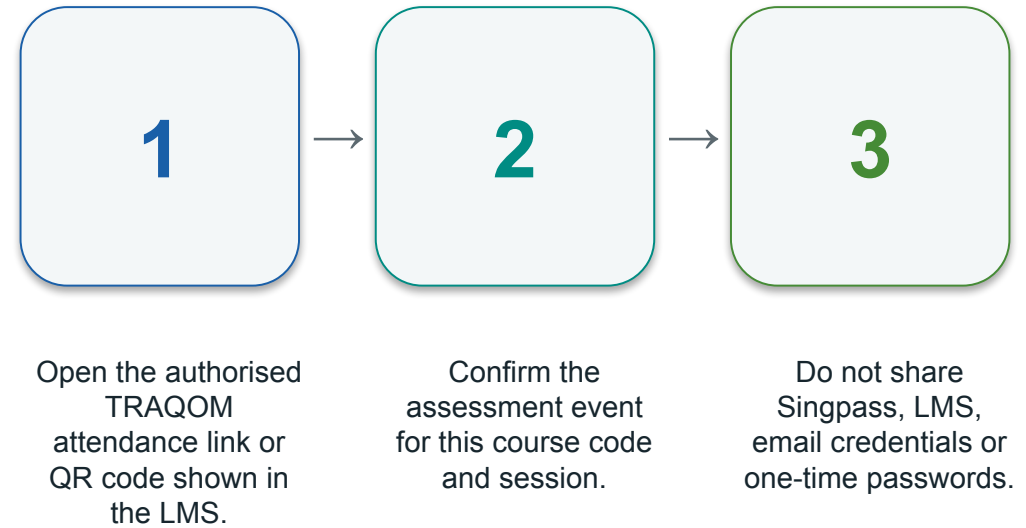


**Assessment sequence: WA-SAQ first, then Practical Performance; follow the candidate instructions and assessor directions.**

# Digital Attendance (Mandatory)

## ANCHOR: TRAQOM · SSG assessment attendance

- Open the authorised TRAQOM attendance link or QR code shown in the LMS.
- Confirm the assessment event for this course code and session.
- Do not share Singpass, LMS, email credentials or one-time passwords.



# Thank You

## ANCHOR: control the root zone, verify the evidence

- Observe the whole water–root–canopy loop.
- Operate to SOP and stop when controls are missing.
- Use measurements and records to improve the next crop cycle.

**1** Observe · whole · water–root–canopy · loop

**2** Operate · stop · when · controls · missing

**3** measurements · records · improve · next · crop · cycle